

Battery Scheduling with Deep Q-Learning

جدولة بطارية باستخدام DQN

A MATLAB custom environment for time-of-use energy management

مخصصة لإدارة الطاقة وفق التعرفة الزمنية MATLAB بيئة

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Laboratory Objectives

- Create a custom battery environment with `rlFunctionEnv`.
- Define observations and discrete actions.
- Implement SOC dynamics, cost, degradation, and penalties.
- Train a DQN agent.
- Compare the learned policy with a no-battery baseline.

أهداف المخبر

- إنشاء بيئة بطارية مخصصة.
- تعريف الملاحظات والأفعال.
- تنفيذ SOC والكلفة والتآكل والعقوبات.
- تدريب وكيل DQN.
- المقارنة مع عدم استخدام البطارية.

1. Requirements

- MATLAB
- Reinforcement Learning Toolbox
- Deep Learning Toolbox

1. المتطلبات

يلزم MATLAB مع Reinforcement Learning Toolbox و Deep Learning Toolbox.

2. Environment

Observation

SOC, load, price, hour.

Actions

-1 charge, 0 idle, +1 discharge.

Episode

24 hourly steps.

Reward

Negative operating cost and penalties.

2. البيئة

تتكون الملاحظة من SOC والحمل والسعر والساعة، والأفعال من الشحن والتوقف والتفريغ، والحلقة يوم واحد.

3. Main MATLAB Script

```

%% Lab 12: Battery Scheduling using DQN
clear; clc; close all;
rng(12);

%% Observation and action specifications
obsInfo = rlNumericSpec([4 1], ...
    LowerLimit=[0;0;0;0], ...
    UpperLimit=[1;2;2;1]);
obsInfo.Name = "battery_observation";

actInfo = rlFiniteSetSpec({-1,0,1});
actInfo.Name = "battery_action";

%% Create custom environment
env = rlFunctionEnv(obsInfo,actInfo, ...
    @batteryStepFunction,@batteryResetFunction);

validateEnvironment(env);

%% Create a default DQN agent
agent = rlDQNAgent(obsInfo,actInfo);

agent.AgentOptions.UseDoubleDQN = true;
agent.AgentOptions.DiscountFactor = 0.99;
agent.AgentOptions.MinibatchSize = 128;
agent.AgentOptions.ExperienceBufferLength = 1e5;
agent.AgentOptions.TargetUpdateFrequency = 4;

agent.AgentOptions.EpsilonGreedyExploration.Epsilon = 1.0;
agent.AgentOptions.EpsilonGreedyExploration.EpsilonMin = 0.05;
agent.AgentOptions.EpsilonGreedyExploration.EpsilonDecay = 1e-4;

%% Training options
trainOpts = rlTrainingOptions( ...
    MaxEpisodes=1200, ...
    MaxStepsPerEpisode=24, ...
    ScoreAveragingWindowLength=50, ...
    StopTrainingCriteria="AverageReward", ...
    StopTrainingValue=-8, ...

```

```

    Verbose=false, ...
    Plots="training-progress");

%% Train
trainingStats = train(agent,env,trainOpts);

%% Simulate one day
simOpts = rlSimulationOptions(MaxSteps=24);
experience = sim(env,agent,simOpts);

obs = experience.Observation.battery_observation.Data;
act = experience.Action.battery_action.Data;
rew = experience.Reward.Data;

soc = squeeze(obs(1,1,:));
actions = squeeze(act);
rewards = squeeze(rew);
hours = 0:numel(actions)-1;

figure;
stairs(0:numel(soc)-1,soc,'LineWidth',1.4);
grid on;
xlabel('Hour'); ylabel('SOC');
title('Battery State of Charge');

figure;
stairs(hours,actions,'LineWidth',1.4);
grid on;
yticks([-1 0 1]);
yticklabels({'Charge','Idle','Discharge'});
xlabel('Hour'); ylabel('Action');
title('DQN Battery Actions');

fprintf('Episode reward = %.3f\n',sum(rewards));

```

3. البرنامج الرئيسي

4. Reset Function

```
function [InitialObservation,Info] = batteryResetFunction()

Info.Load = [2.2 2.0 1.9 1.8 1.9 2.3 ...
             3.0 3.8 4.2 4.0 3.7 3.5 ...
             3.4 3.6 4.0 4.8 5.8 6.5 ...
             6.1 5.4 4.7 4.0 3.2 2.6]';

Info.Price = [0.18*ones(6,1); ...
             0.25*ones(10,1); ...
             0.42*ones(5,1); ...
             0.23*ones(3,1)];

Info.Capacity = 10;           % kWh
Info.Pmax = 2.5;              % kW
Info.EtaCharge = 0.95;
Info.EtaDischarge = 0.95;
Info.SOCmin = 0.10;
Info.SOCmax = 0.90;

Info.Hour = 1;
Info.SOC = 0.50;
Info.TotalCost = 0;

InitialObservation = makeObservation(Info);
end
```

4. دالة إعادة الضبط

5. Step Function

```

function [NextObservation,Reward,IsDone,Info] = ...
    batteryStepFunction(Action,Info)

action = double(Action);
h = Info.Hour;
loadPower = Info.Load(h);
price = Info.Price(h);

% -1 charge, 0 idle, +1 discharge
PbattCommand = action*Info.Pmax;
socOld = Info.SOC;
violation = 0;

if PbattCommand < 0
    Pcharge = -PbattCommand;
    socNew = socOld + ...
        Info.EtaCharge*Pcharge/Info.Capacity;
elseif PbattCommand > 0
    Pdischarge = PbattCommand;
    socNew = socOld - ...
        Pdischarge/(Info.EtaDischarge*Info.Capacity);
else
    socNew = socOld;
end

if socNew > Info.SOCmax
    violation = socNew-Info.SOCmax;
    socNew = Info.SOCmax;
elseif socNew < Info.SOCmin
    violation = Info.SOCmin-socNew;
    socNew = Info.SOCmin;
end

deltaSOC = socNew-socOld;

if deltaSOC >= 0
    gridBatteryPower = ...
        deltaSOC*Info.Capacity/Info.EtaCharge;
else

```



```

    deliveredPower = ...
        -deltaSOC*Info.Capacity*Info.EtaDischarge;
    gridBatteryPower = -deliveredPower;
end

gridPower = max(0,loadPower+gridBatteryPower);
energyCost = price*gridPower;
degradationCost = 0.015*abs(gridBatteryPower);

Reward = -energyCost ...
        -degradationCost ...
        -30*violation;

Info.SOC = socNew;
Info.TotalCost = Info.TotalCost+energyCost;
Info.Hour = Info.Hour+1;

IsDone = Info.Hour > 24;

if IsDone
    terminalPenalty = 3*abs(Info.SOC-0.50);
    Reward = Reward-terminalPenalty;
    Info.Hour = 24;
end

NextObservation = makeObservation(Info);
end

```

5. دالة الخطوة

تحدث حالة الشحن، وتفرض الحدود، وتحسب قدرة الشبكة والكلفة والتآكل والمكافأة.

6. Observation Helper

```

function observation = makeObservation(Info)
h = min(Info.Hour,24);

loadNorm = Info.Load(h)/max(Info.Load);
priceNorm = Info.Price(h)/max(Info.Price);
hourNorm = (h-1)/23;

observation = [
    Info.SOC
    loadNorm
    priceNorm
    hourNorm
];
end

```

6. بناء الملاحظة

تُطبع قيم الحمل والسعر والساعة وتُجمع مع SOC في متجه من أربعة عناصر.

7. Reward

$$r_t = -Cost_t - Degradation_t - 30 \cdot SOCViolation_t$$

A final-SOC penalty prevents the agent from emptying the battery at the end of every episode.

7. المكافأة

8. Baseline

```
[~,baseInfo] = batteryResetFunction();  
baselineCost = sum(baseInfo.Load.*baseInfo.Price);  
fprintf('No-battery daily cost = %.2f EUR\n',baselineCost);
```

Judge the trained policy against a simple baseline, not only by episode reward.

8. خط الأساس

احسب كلفة شراء الحمل كاملاً دون بطارية ثم قارنها بسياسة DQN.

9. Expected Behavior

- Charge during low-price hours.
- Discharge during expensive evening hours.
- Avoid SOC violations.
- Preserve final SOC.

Training is stochastic. Stopping thresholds may need adjustment when reward scaling changes.

9. السلوك المتوقع

يُتوقع الشحن في الساعات الرخيصة والتفريغ في الذروة مع احترام الحدود والمحافظة على SOC نهائية مناسبة.

10. Student Experiments

1. Change capacity to 5 and 20 kWh.
2. Use five action levels.
3. Add PV generation.
4. Add a demand-charge penalty.
5. Randomize load and price at reset.
6. Compare with a rule-based controller.

10. تجارب الطالب

1. غيّر سعة البطارية.
2. استخدم خمسة مستويات قدرة.
3. أضف PV.
4. أضف عقوبة للذروة.
5. غيّر الحمل والسعر عشوائيًا.
6. قارن بتحكم قائم على قواعد.

11. Mini Project

Microgrid extension: Add PV, a diesel generator, grid-import limits, and load shedding. Balance cost, emissions, reliability, and degradation.

11. مشروع مصغر

وسّع البيئة إلى شبكة مصغرة تضم PV ومولد ديزل وحدود استيراد وفصل أحمال.

12. Laboratory Report

- Define the MDP.
- Explain SOC dynamics.
- Plot training reward, SOC, and actions.
- Compare cost with the baseline.
- Report violations.
- Discuss reward sensitivity and safety.

12. تقرير المخبر

يتضمن تعريف MDP ومعادلات SOC والرسوم والمقارنة مع خط الأساس والمخالفات وتحليل المكافأة والأمان.